Quadratics: Expanding & Factorising

Practice test 8 · 40 marks · Higher tier · non-calculator

Name: _____ Date: _____ Score: _____ / 40

Section A — Expanding double brackets

A1. Expand and simplify (x + 10)(x + 3)

[2 marks]

A2. Expand and simplify (x – 4)(x + 13)

[2 marks]

A3. Expand and simplify (x – 2y)(x + 5y)

[2 marks]

A4. Expand and simplify $3(x + 2)^2 – (x + 4)(x – 1)$

[2 marks]

Section B — Expanding three brackets

B1. Expand and simplify (x – 4)(x + 1)(x + 3)

[3 marks]

B2. Expand and simplify $(2x + 1)^3$

[3 marks]

Section C — Factorising $x^2 + bx + c$ and the difference of two squares

C1. Factorise $x^2 + 10x + 21$

[2 marks]

C2. Factorise $x^2 – 9x – 36$

[2 marks]

C3. Factorise $25x^2 – 64$

[1 mark]

C4. Factorise fully $2x^3 – 18x$

[2 marks]

Section D — Factorising $ax^2 + bx + c$

D1. Factorise $2x^2 + 3x – 20$

[2 marks]

D2. Factorise $3x^2 – 17x + 10$

[2 marks]

D3. Factorise $6x^2 - 13x - 5$

[2 marks]

Section E — Using factorising

E1. Solve $x(x + 4) = 21$

[3 marks]

E2. Simplify fully

$$\frac{3x^2 - 12}{x^2 + x - 6}$$

[3 marks]

Section F — Problem solving & proof

F1. Write $3x^2 - 12x + 5$ in the form $a(x + b)^2 + c$

[3 marks]

F2. A square lawn has sides of x metres. A path 2 m wide goes all the way round the outside of the lawn. Show that the area of the path is $(8x + 16)$ m^2.

[3 marks]

F3. Without a calculator, work out $12.5^2 - 7.5^2$

[1 mark]